

A General Model of Causal Dissonance, Directed Inquiry, and the Return to Bayesian Competence

Research note

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Abstract

This note generalizes the four-state laboratory to an arbitrary finite number of agents, a general finite state space, and stochastic consequences. The architecture follows the subjective-rationality tradition: each agent adopts a causal theory, combines it with beliefs and conjectures to choose a committed policy, and the profile of policies determines everyone's consequences through an objective causal system. All market and environmental detail is absorbed into that system; the front end—awareness, dissonance, inquiry, insight, and the reconstruction of belief—parallels the laboratory. The organizing object is the chain

dissonance $\longrightarrow$ inquiry $\longrightarrow$ refined awareness $\longrightarrow$ new models $\longrightarrow$ Bayesian judgment,

the process by which an agent whose causal language has failed regains the ability to be a Bayesian. Theories are assignments of elementary causal mechanisms to awareness cells; dissonance is the statistical rejection of every expressible assignment; inquiry is a costly, delabeled election; insight is a fallible refinement of the partition; and the bridge prior restores a Bayesian posture on the expanded language. A self-confirming unawareness equilibrium is defined not as a strategic rest point but as the configuration in which every agent permanently inhabits a Bayesian world: models fit, questions are dead, and inquiry is rationally unpriced. Convergence to such configurations, their possible falsity off the exposed path, and the persistence of performance differences follow from the economics of the process rather than from an equilibrium postulate.

1 Purpose, fulcrum, and relation to the laboratory

This note supplies the general canvas for the ideas made concrete in the four-state laboratory (Ryall, 2026a). Its subject is the process by which a strategic agent whose causal representation has failed regains the capacity for ordinary Bayesian reasoning: experience rejects every expressible theory; the rejection localizes a question; costly inquiry may produce a previously inexpressible distinction; the expanded language admits new theories; and belief formation resumes. Equilibrium concepts appear only at the end, and only in a subordinate role: a self-confirming unawareness equilibrium describes the configuration in which this process has permanently terminated—every agent lives in a Bayesian world and has no represented reason to leave it. Persistence of theory heterogeneity and of performance differences is derived from the economics of the process: from what inquiry costs, what questions are worth, which experiences the play of the game supplies, and where refinement paths lead.

The back end follows the subjective-rationality tradition of Kalai and Lehrer (1993b) and Ryall (2003). A finite set of agents adopt causal theories; theories, beliefs, and conjectures determine

committed policies; and the policy profile determines stochastic consequences for every agent through an objective causal system. Everything about markets, customers, and technology is absorbed into that system. The note adds to this tradition the front end: the agents' *language* for the system—which distinctions they can represent—is endogenous, and the fixed-hypothesis-space model is recovered as the special case in which awareness never changes.

Two disciplines are maintained throughout. First, causality: the objective system specifies the distribution of consequences that *would* result from every action profile at every state, whether or not that profile is ever played. Agents' actions cause their consequences; theories are causal models of that relation; and the gap between on-path evidence and counterfactual content is what the identification results bite on. No structural-causal-model formalism is imposed—mechanisms are reduced-form causal objects—but detailed micro-models can generate them in examples, as the laboratory's winner-take-all customer does. Second, the no-baked-in checklist of the companion memorandum: nothing behind an agent's awareness frontier may carry semantic content or objective payoff information into pre-insight deliberation.

2 Notation

Symbol	Meaning
$N = \{1, \dots, n\}$	Set of agents.
S	Finite state space; s a state.
$\rho \in \Delta(S)$	State distribution (i.i.d. draws; full support).
$A_i, A = \prod_i A_i$	Agent i 's finite action set; action profiles a .
$X_i, X = \prod_i X_i$	Agent i 's finite consequence space; profiles x .
$u_i : X_i \rightarrow \mathbb{R}$	Agent i 's payoff, a function of its consequence.
Θ	Finite set of elementary causal mechanisms $\theta : A \rightarrow \Delta(X)$.
$\tau : S \rightarrow \Theta$	Objective mechanism assignment.
$F(\cdot \mid s, a)$	Objective causal system: $F(\cdot \mid s, a) = \tau(s)(a)$.
$\Pi(S), \mathcal{P}_i, C$	Partitions of S ; agent i 's awareness partition; a cell.
$M(\mathcal{P})$	Expressible theories: assignments $m : \mathcal{P} \rightarrow \Theta$.
$\mu_i \in \Delta(M(\mathcal{P}_i))$	Agent i 's belief over expressible theories.
$q_i : \mathcal{P}_i \rightarrow \Delta(A_{-i})$	Agent i 's conjecture about rival play.
$\sigma_i : \mathcal{P}_i \rightarrow A_i$	Agent i 's committed policy.
g, T	Generation index; episodes per generation.
r_i	Agent i 's deliberative record of retained episode tokens.
$n_i(C, a), \hat{p}_i(\cdot \mid C, a)$	Observation count and empirical conditional at (C, a) .
$p_i^*(\cdot \mid C, a)$	On-path conditional induced by (τ, ρ, σ) .
$\eta, w(\cdot), \bar{n}$	Test tolerance, slack sequence, minimum count.
$\hat{M}_i(\mathcal{P}, r_i)$	Test-active set of expressible theories.
$\mathcal{R}(\mathcal{P}, r_i)$	Minimally restorative refinements.
κ	Complexity ordering on refinements (technology primitive).
ϕ_i	Representation-producing technology.
$\varepsilon_i, \gamma_i, \lambda_i$	Inquiry election, cost, and delabeled success weight.
$B_i(r_i)$	Retained-record leverage of inquiry.
$\mathcal{T}, \nu$	Candidate assignments in a reflective assessment; reflective prior.
W, G	Selection-adjusted warrant; warrant gap.

In the four-state laboratory, customer states were written x ; here s denotes states and x denotes consequences, matching the convention that consequences are what agents see. [Section 10](#) gives the full mapping.

3 Primitives

3.1 Agents, states, actions, consequences

There are n agents, $N = \{1, \dots, n\}$. Time is divided into generations $g = 1, 2, \dots$, each consisting of T episodes. In every episode a state $s \in S$ is drawn independently from a full-support distribution $\rho \in \Delta(S)$; each agent takes an action $a_i \in A_i$ (through the committed policy defined below); and a consequence profile $x = (x_1, \dots, x_n) \in X$ is realized. Agent i receives payoff $u_i(x_i)$. The

consequence x_i is everything agent i obtains from the episode’s outcome—profit, sales, observed reactions—and payoffs are functions of consequences rather than separate primitives.

3.2 Mechanisms and the objective causal system

Definition 3.1 (Elementary mechanisms and the objective system). *A finite set Θ of elementary causal mechanisms is given, each a map $\theta : A \rightarrow \Delta(X)$. The objective assignment is a function $\tau : S \rightarrow \Theta$, and the objective causal system is*

$$F(\cdot \mid s, a) = \tau(s)(a) \in \Delta(X). \quad (1)$$

Two states are causally equivalent if they carry the same mechanism. Write $\mathcal{P}^\tau$ for the partition of S into causal-equivalence classes and $\text{marg}_i \theta(a)$ for the X_i -marginal of $\theta(a)$.

Remark 3.2 (Causal reading). *The system F is a response function, not a statistical summary: it specifies the distribution of consequences that would result from every action profile at every state, including profiles never played. Agents’ actions, together with the state, cause consequences; a mechanism is the reduced form of whatever detailed model—a structural causal model of customers, a market microstructure, a production process—underlies that relation, and examples may exhibit such micro-foundations explicitly. Three consequences of the causal reading recur below. First, theories (assignments of mechanisms to cells) are interventional objects: a theory makes predictions at unplayed profiles. Second, on-path data constrain F only at visited state–profile pairs, so observationally equivalent systems with different counterfactual content are generic; that is the engine of the nonidentification results. Third, the on-path conditional distribution of consequences given an awareness cell is a selection object—it mixes mechanisms according to which states co-occur with which play—and must not be conflated with any mechanism’s own conditional.*

Remark 3.3 (Why theories are mechanism assignments). *If expressible theories were arbitrary conditional distributions over consequences per cell, any within-cell mixture would itself be expressible and no experience could reject the whole class: coarseness alone cannot produce dissonance in a saturated model space. The laboratory avoided this because its theories were the structured labels $\{0, 1, n\}$. The general counterpart is Θ : an awareness cell is the agent’s implicit claim that its states are causally alike, an expressible theory assigns one elementary mechanism to each cell, and dissonance—defined below—is statistical evidence of mechanism heterogeneity inside a represented category. What insight produces is the distinction that splits the category; the mechanism vocabulary Θ itself is known to all agents from the outset, exactly as the labels were in the laboratory. Expansion of the mechanism vocabulary is a further, deeper dimension of awareness growth deliberately left outside this paper.*

3.3 Awareness, theories, conjectures, and policies

Definition 3.4 (Awareness and expressible theories). *Agent i ’s awareness partition is $\mathcal{P}_i \in \Pi(S)$. Every consciously available object—theory, belief, conjecture, policy, question—must be measurable with respect to $\mathcal{P}_i$. The expressible theories are the cell-constant mechanism assignments*

$$M(\mathcal{P}_i) = \{m : \mathcal{P}_i \rightarrow \Theta\}, \quad (2)$$

with m predicting $x_i \sim \text{marg}_i m(C)(a)$ at cell C under profile a . The objective assignment τ is expressible for agent i if and only if τ is constant on every cell of $\mathcal{P}_i$, that is, if and only if $\mathcal{P}_i$ refines $\mathcal{P}^\tau$; call such a partition sufficient for τ .

Agent i holds a belief $\mu_i \in \Delta(M(\mathcal{P}_i))$ over expressible theories and a conjecture $q_i : \mathcal{P}_i \rightarrow \Delta(A_{-i})$ about rival play, both $\mathcal{P}_i$ -measurable. Theories and conjectures are different kinds of objects and are deliberately separated: a theory is a causal claim about the objective system; a conjecture is a strategic claim about what other agents will do. Dissonance, defined below, tests theories, not conjectures.

Assumption 3.5 (Observation and retention). *At the end of each episode, agent i observes the triple $(C_{\mathcal{P}_i}(s), a, x_i)$: its cell, the realized action profile, and its own consequence. Episodes are retained as tokens in a deliberative record r_i and can be retrospectively recoded after the agent acquires a new distinction; before acquisition, neither belief nor action can condition on that distinction. The analyst-level history additionally records s , so that a newly produced representation can recode the tokens.*

Observability of the full profile is the laboratory’s convention and keeps the separation of causal from strategic error clean; a variant in which components of a are folded into the consequence x_i (so that x_i is literally everything seen) changes bookkeeping but nothing conceptual.

Assumption 3.6 (Commitment and subjective optimality). *At the start of each generation, agent i commits to a policy $\sigma_i : \mathcal{P}_i \rightarrow A_i$ maximizing subjective expected payoff,*

$$\sigma_i(C) \in \arg \max_{a_i \in A_i} \sum_{m \in M(\mathcal{P}_i)} \mu_i(m) \sum_{a_{-i} \in A_{-i}} q_i(a_{-i} \mid C) \mathbb{E}_{x \sim m(C)(a_i, a_{-i})} [u_i(x_i)], \quad (3)$$

holds the policy for the generation’s T episodes, and updates only between generations. When dissonance has made μ_i undefined and no new representation has been produced, the previously committed policy is carried forward. Agents evaluate one generation ahead; no discounting is imposed.

Commitment is where the statistical structure and the economics meet: a test needs a stable data-generating configuration, and a generation of committed play supplies exactly that. What the laboratory assumed as a batching device is, in the stochastic model, the natural granularity of learning.

3.4 On-path conditionals

Fix a generation with committed profile $\sigma = (\sigma_1, \dots, \sigma_n)$. Each state s receives the deterministic profile $a(s) = (\sigma_j(C_{\mathcal{P}_j}(s)))_{j \in N}$. Agent i ’s *on-path conditional* at an observation pair (C, a) with $S(C, a) = \{s \in C : a(s) = a\} \neq \emptyset$ is

$$p_i^*(\cdot \mid C, a) = \sum_{s \in S(C, a)} \rho(s \mid S(C, a)) \text{ marg}_i \tau(s)(a). \quad (4)$$

This is the selection object of [remark 3.2](#): rivals with finer partitions condition on distinctions i cannot see, so the within-cell state mixture can vary with the profile. It is precisely this dependence—differentiated play by others inside one’s own cell—that generates the contrasts on which dissonance feeds.

4 The Bayesian layer and its boundary

Within fixed awareness, learning is ordinary. Observations representable in the current language update μ_i by Bayes’ rule; conjectures are updated empirically from observed play. This layer nests the fixed-vocabulary subjective-rationality models: if no agent’s awareness ever changes, the model is a repeated interaction among subjectively rational Bayesian learners in the sense of [Kalai and Lehrer \(1993b\)](#) and [Ryall \(2003\)](#).

Proposition 4.1 (No Bayesian emergence). *Bayesian conditioning on any event measurable with respect to the algebra generated by $\mathcal{P}_i$ yields a posterior on $\Delta(M(\mathcal{P}_i))$. No sequence of such updates makes a theory outside $M(\mathcal{P}_i)$ expressible or a distinction finer than $\mathcal{P}_i$ usable in a policy.*

Proof. Conditioning reweights a distribution on its domain and cannot enlarge the domain; every element of $M(\mathcal{P}_i)$ is cell-constant, so every posterior mixture and every policy optimal against it is $\mathcal{P}_i$ -measurable. $\square$

Remark 4.2 (The posterior is self-sealing). *The stochastic setting adds a sharper point. Suppose every expressible theory is false—the cell mixes mechanisms, so no cell-constant assignment matches the on-path conditionals. A full-support Bayesian posterior does not signal this: by the logic of misspecified Bayesian learning (Berk, 1966), the posterior concentrates on the expressible theories closest to the truth in Kullback–Leibler divergence, and the agent becomes increasingly confident in a wrong theory. Posterior mass always sums to one; model failure is not an event in the agent’s algebra. Detecting that the language itself is broken therefore requires a faculty outside the posterior—the consistency test of the next section. This is the formal counterpart of separating understanding from judgment: updating ranks the answers one can express, while judgment asks whether any expressible answer is adequate.*

5 Tests, active sets, and statistical dissonance

5.1 The consistency test

Definition 5.1 (Consistency test and active set). *Fix a tolerance $\eta \geq 0$, a minimum count $\bar{n} \geq 1$, and a nonincreasing slack sequence $w(\cdot) \rightarrow 0$. Given record r_i with counts $n_i(C, a)$ and empirical conditionals $\hat{p}_i(\cdot \mid C, a)$, theory $m \in M(\mathcal{P}_i)$ passes if*

$$\text{TV}(\hat{p}_i(\cdot \mid C, a), \text{marg}_i m(C)(a)) \leq \eta + w(n_i(C, a)) \quad \text{for every } (C, a) \text{ with } n_i(C, a) \geq \bar{n}. \quad (5)$$

The active set $\hat{M}_i(\mathcal{P}_i, r_i)$ is the set of passing theories. Call m η -adequate under a committed profile if $\text{TV}(\hat{p}_i^(\cdot \mid C, a), \text{marg}_i m(C)(a)) \leq \eta$ at every visited pair, and η -inadequate if the inequality fails strictly at some visited pair.*

Lemma 5.2 (Test consistency). *Fix a committed profile played in all generations and suppose the tolerance is generic in the sense that no theory’s distance to an on-path conditional equals η exactly. As generations accumulate, almost surely: every η -adequate theory is eventually always in the active set, and every η -inadequate theory is eventually never in it.*

Proof. Visited pairs have $n_i(C, a) \rightarrow \infty$ almost surely under i.i.d. full-support states, and $\hat{p}_i(\cdot \mid C, a) \rightarrow p_i^*(\cdot \mid C, a)$ by the strong law. The triangle inequality then places $\text{TV}(\hat{p}_i, \text{marg}_i m(C)(a))$ within any ϵ of $\text{TV}(p_i^*, \text{marg}_i m(C)(a))$ eventually; genericity converts the strict inequalities into eventual acceptance and rejection, and the slack term absorbs finite-sample fluctuation. $\square$

The deterministic laboratory is the degenerate instance $\eta = 0$, $\bar{n} = 1$, $w \equiv 0$ with deterministic mechanisms: passing is exact consistency, and the active set is the laboratory’s.

Definition 5.3 (Causal dissonance). *Agent i experiences causal dissonance at the end of a generation if $\hat{M}_i(\mathcal{P}_i, r_i) = \emptyset$: every expressible theory is rejected. The expressible set does not disappear; no element of it accounts for the record. The immediate question has a determinate defect but not a determinate answer: what unrepresented difference among the pooled states explains the heterogeneity of their consequences?*

5.2 Exposure

Dissonance requires more than falsity; the play of the game must make the falsity visible.

Definition 5.4 (Exposure). *Under a committed profile, awareness $\mathcal{P}_i$ is exposed if every $m \in M(\mathcal{P}_i)$ is η -inadequate. A committed profile is concealing for agent i if some expressible theory is η -adequate under it despite being objectively false—false, that is, as a claim about τ at some state its cells cover.*

Proposition 5.5 (Exposure and dissonance). *Under a fixed committed profile, agent i experiences dissonance from some generation onward almost surely if and only if $\mathcal{P}_i$ is exposed.*

Proof. Immediate from lemma 5.2: exposure means every theory is eventually rejected; its failure means some theory is eventually always accepted. $\square$

Exposure generalizes the laboratory’s revelation lemma and its demand-side extension in one object. There, pooled play under forgiving demand concealed the row law from column believers; harsh demand or differentiated rivals exposed it. Here, whether heterogeneity inside a cell surfaces depends on which profiles are visited (strategic differentiation inside one’s cells) and on how sharply mechanisms differ at the visited profiles (the environment’s harshness). Both sources of diagnosticity reappear in the dynamics of section 9.

6 Inquiry: restoration, direction, and fallibility

6.1 Restorative refinements

For $\mathcal{P}, \mathcal{P}' \in \Pi(S)$, write $\mathcal{P} \prec \mathcal{P}'$ when $\mathcal{P}'$ strictly refines $\mathcal{P}$. After a refinement, the retained record is recoded (assumption 3.5) and the test is re-applied with the recoded counts.

Definition 6.1 (Restorative refinement). *$\mathcal{P}'$ is restorative at $(\mathcal{P}_i, r_i)$ if $\mathcal{P}_i \prec \mathcal{P}'$ and $\widehat{M}_i(\mathcal{P}', r_i) \neq \emptyset$ under the recoded record. It is minimally restorative if no restorative $\mathcal{P}''$ satisfies $\mathcal{P}_i \prec \mathcal{P}'' \prec \mathcal{P}'$. Write $\mathcal{R}(\mathcal{P}_i, r_i)$ for the set of minimally restorative refinements.*

Proposition 6.2 (Cellwise coherence). *Call a cell C' of a refinement coherent if some single mechanism passes the test on C' ’s recoded observations. Then $\widehat{M}_i(\mathcal{P}', r_i) \neq \emptyset$ if and only if every cell of $\mathcal{P}'$ is coherent.*

Proof. Theories factor across cells— $M(\mathcal{P}') = \Theta^{\mathcal{P}'}$ and the test constraints in equation (5) are per cell—so a passing assignment exists exactly when a passing mechanism can be chosen cell by cell. $\square$

Coherence generalizes the laboratory’s conflict-free condition. There, incoherence was a binary relation between two labeled episodes and restorative partitions were proper colorings of a conflict graph; here, incoherence is in general a property of sets of tokens (a mixture of three mechanisms may be incompatible with every single mechanism even though every pair of tokens is individually accommodatable), so restoration is a hypergraph-coloring problem. Two structural facts survive unchanged: restoration is always possible (the partition into singletons recodes every token to its own state, where a single mechanism always fits a single-state record), and minimal restorations are generally multiple—the record directs inquiry without determining its output.

6.2 The technology

Definition 6.3 (Representation-producing technology). *A representation-producing technology for agent i is a (possibly randomized) map ϕ_i from $(\mathcal{P}_i, r_i)$ to $\Pi(S)$ with $\phi_i(\mathcal{P}_i, r_i) \in \{\mathcal{P}_i\} \cup \mathcal{R}(\mathcal{P}_i, r_i)$: it fails, or it succeeds and produces a minimally restorative refinement. It is truth-blind: its output distribution depends only on $(\mathcal{P}_i, r_i)$, never directly on τ . A complexity ordering κ on refinements—the analyst’s model of which conceptual products the agent’s comparison capacities can generate—may restrict the range further, as the coordinate-complexity filter did in the laboratory; κ is a primitive of the technology, not an object of the agent’s optimization.*

Assumption 6.4 (Dissonance-triggered, costly, elective production). *ϕ_i operates only when agent i experiences dissonance and elects inquiry, $\varepsilon_i = 1$, at cost $\gamma_i > 0$. Agent i holds a delabeled success weight $\lambda_i \in (0, 1]$: a subjective probability that an elected episode produces a usable refinement, defined on the event of success alone and not on the identities of possible outputs. On failure or abstention the committed policy is carried forward. Awareness is cumulative: acquired distinctions are never lost, though theories formulated on them may be rejected.*

6.3 Nonidentification and the price of robustness

Theorem 6.5 (Observational nonidentification). *Let $\tau \neq \tau'$ be objective assignments that induce the same distribution over agent i ’s records under the committed profiles played. Then any truth-blind technology and any truth-blind inquiry-and-formulation rule has the same output distribution in both worlds. If, in addition, no single theory is η -adequate under both worlds at the visited pairs, then no truth-blind rule selects an adequate formulation with probability one in both worlds.*

Proof. The rule’s input is the record, whose distribution is identical by hypothesis; its output distribution is therefore identical. If the adequate sets are disjoint, any output is inadequate in at least one world with the corresponding probability. $\square$

Proposition 6.6 (Robust sufficiency). *A partition sufficient for every assignment in a candidate set $\mathcal{T}$ must refine $\bigvee_{\tau' \in \mathcal{T}} \mathcal{P}^{\tau'}$, the join of the candidates’ causal-equivalence partitions. Representations guaranteed to accommodate all observationally equivalent worlds are accordingly finer than any single world requires, and a technology that economizes on distinctions must accept fallibility.*

Proof. Sufficiency for τ' is refinement of $\mathcal{P}^{\tau'}$ (definition 3.4); refinement of each is refinement of the join. $\square$

In the laboratory, the join of the row and column worlds was the singleton partition; proposition 6.6 is that result in general form.

7 The economics of inquiry

The election ε_i must be priced at exactly the moment Bayes is silent: the posterior is undefined at dissonance (definition 5.3), and the candidate answers are not represented objects (definition 6.3). The laboratory’s solution generalizes with one qualification.

Definition 7.1 (Pinned distinctions). *Token sets $K, K' \subseteq r_i$ are mutually incoherent if no single mechanism passes the test on $K \cup K'$. A pinned distinction is a separation of mutually incoherent token sets. Every restorative refinement separates every pinned distinction—a cell containing both sets would be incoherent (proposition 6.2)—so conditioning future play on a pinned distinction is a capability that any successful inquiry outcome delivers.*

Definition 7.2 (Retained-record leverage). *Under the subjective recurrence premise of the laboratory—the agent evaluates one generation ahead as if its retained record recurs token for token—let $B_i(r_i)$ be the largest per-generation payoff improvement over the carried-forward policy that is achievable by a contingent rule measurable with respect to pinned distinctions, computable from the record together with the known structure (Θ, u_i) , and robust to every rival behavior consistent with the record. Set $B_i = 0$ when there is no dissonance.*

Proposition 7.3 (Answer invariance and participation). *$B_i(r_i)$ is computable from the deliberative record alone and is identical across all elements of $\mathcal{R}(\mathcal{P}_i, r_i)$: the agent prices the question without a menu of answers. Electing inquiry has represented value $\lambda_i B_i(r_i) - \gamma_i$, so agent i elects if and only if $\lambda_i B_i(r_i) \geq \gamma_i$. Moreover $B_i > 0$ only if some pinned distinction changes the robustly optimal action at some recurrence: questions whose every admissible answer supports the current action are worthless at any positive cost, however much model multiplicity remains.*

Proof. Measurability with respect to pinned distinctions is deliverable by every restorative refinement (definition 7.1), so the maximand in definition 7.2 does not depend on which refinement arrives; the value of electing follows as in the laboratory, with the extrapolated component of any full formulation assigned no represented value at the election. If no pinned distinction changes the robustly optimal action, every admissible contingent rule is payoff-equivalent to the carried-forward policy on the recurrence record, so the maximum is zero. $\square$

Remark 7.4 (Exactness is special; opacity suppresses inquiry). *In the laboratory at $\beta = 1$, the constant-sum two-action structure made the unit gain of a pinned separation exactly one-half against every rival action. Generally the robust bound is not exact, and—more interestingly—it can be zero even when refinement would objectively pay, because computing forgone payoffs from the record requires the consequence space to be counterfactually transparent: rich enough that observed consequences, together with the known mechanism family, identify what other own-actions would have yielded. Winner-take-all markets are transparent; environments whose consequences conceal forgone values depress B_i toward zero and hence rationally suppress inquiry. How much an environment lets agents price their own questions is itself a dimension of diagnosticity, alongside exposure.*

8 The pivot: formulation, warrant, and the return to Bayes

Successful production yields $\mathcal{P}'_i$; formulation selects $m' \in \widehat{M}_i(\mathcal{P}'_i, r_i)$; and the agent forms a full-support bridge prior $\mu_i^+ \in \Delta(M(\mathcal{P}'_i))$, from which ordinary updating resumes. Nothing in Bayesian coherence determines μ_i^+ from the pre-dissonance belief—the domains differ—and the renormalization of μ_i^+ on the active set is *representation-relative coherence*, not warrant. The laboratory’s accounting of the difference generalizes.

Definition 8.1 (Reflective assessment, warrant, and the gap). *A reflective assessment problem is a pair $(\mathcal{T}, \nu)$: a finite set of candidate objective assignments with a full-support reflective prior. Given the record and a successful production-and-formulation event, the selection-adjusted warrant W of $(\mathcal{P}'_i, m')$ is the ν -posterior probability that m' is objectively correct—that $\tau'(s) = m'(C_{\mathcal{P}'_i}(s))$ for all s —conditional on the record and the production event. The warrant gap G is the representation-relative coherence probability of m' minus W .*

Theorem 8.2 (The insight is not evidence). *If the technology is truth-blind, conditioning on the production event adds nothing to the record: W equals the ordinary reflective posterior of m' ’s correctness given the record alone. Moreover, post-restoration fit is guaranteed by the restoration property, so passing the test carries no information beyond the record either. Among candidates in $\mathcal{T}$ that induce the same record distribution, prior odds are preserved exactly.*

Proof. The reflective posterior on $\tau' \in \mathcal{T}$ is proportional to $\nu(\tau')$ times the record's likelihood under τ' times the probability of the production event given the record; truth blindness makes the last factor a function of the record alone, constant across $\mathcal{T}$, so it cancels. Fit is a deterministic function of record and output (definition 6.1), hence likelihood one in every world consistent with the event. The final claim is the cancellation applied to candidates with identical record likelihoods. $\square$

Unlike the deterministic laboratory, the record itself is generally informative among candidates—stochastic likelihoods differ—so warrant need not equal the prior. The theorem's content is the location of the boundary: everything evidential is in the record; nothing evidential is in the arrival or the retrospective success of the insight. If production were correlated with the objective assignment—social transmission from an informed source—the event would carry information, and the adjustment could move confidence in either direction.

Corollary 8.3 (Self-audit requires sufficiency for the candidates). *The assessment $(\mathcal{T}, \nu)$ is expressible by agent i only if $\mathcal{P}_i$ is sufficient for every $\tau' \in \mathcal{T}$, hence only if $\mathcal{P}_i$ refines the join of proposition 6.6. An agent holding a minimal restoration therefore generally cannot formulate the comparison that measures its own warrant gap; short of that refinement, warrant must be supplied externally—by fresh evidence whose likelihoods differ across candidates, by an assessor with broader awareness, or by a future self.*

Proof. Entertaining τ' as a hypothesis requires expressing it, which is sufficiency (definition 3.4); sufficiency for all candidates is refinement of the join. $\square$

The pivot is now complete, and it is the note's fulcrum: dissonance broke the agent's Bayesian competence; inquiry produced a distinction; formulation and the bridge prior rebuilt a probability space; and judgment—fresh evidence with differential likelihoods, honestly accounted—resumes as ordinary updating on the expanded language. The agent has returned to the Bayesian world, carrying a warrant gap of known location and, in general, unknown (to it) size.

9 Dynamics: when everyone lives in a Bayesian world

The chain now runs inside a population: theories determine policies, policies determine which conditionals are visited, visited conditionals determine dissonance and leverage, and refinements feed back into theories. This section characterizes where the process comes to rest. The interesting object is not the rest configuration as an equilibrium but what it asserts epistemically: *a state of the population in which every agent permanently inhabits a Bayesian world*—models fit, questions are dead, and no represented calculation prices further inquiry above its cost.

Lemma 9.1 (Finitely many refinements). *Each successful production strictly increases the number of cells of some agent's partition, so along any path there are at most $n(|S| - 1)$ successful productions. Almost surely there is a last one.*

Proof. A strict refinement of a partition of a finite set adds at least one cell, and cell counts are bounded by $|S|$; cumulative awareness (assumption 6.4) rules out coarsening. $\square$

Definition 9.2 (Self-confirming unawareness equilibrium). *A profile $(\mathcal{P}_i, \mu_i, q_i, \sigma_i)_{i \in N}$ is a self-confirming unawareness equilibrium (SCUE) if, under the induced play,*

- (i) *each σ_i is subjectively optimal (equation (3));*
- (ii) *each conjecture q_i agrees with the play induced by σ_{-i} at every on-path cell;*

- (iii) *each agent's active set is almost surely eventually nonempty and each μ_i is supported on η -adequate theories: no dissonance occurs; and*
- (iv) *no agent elects inquiry: $\lambda_i B_i < \gamma_i$, which holds in particular whenever $B_i = 0$.*

Conditions (i)–(ii) are subjective rationality with confirmed conjectures; they are the self-confirming-equilibrium conditions of the fixed-language tradition. Conditions (iii)–(iv) are the new, epistemic ones: the language survives its own evidence, and the question has no represented value. A SCUE with every $\mathcal{P}_i$ sufficient for τ and beliefs concentrated near truth is simply a self-confirming equilibrium; the concept's interest lies in the other cases.

Remark 9.3 (The grain of truth, restored endogenously). *Convergence results in the rational-learning tradition require a grain of truth: the true environment must be absolutely continuous with respect to beliefs (Kalai and Lehrer, 1993a). Under unawareness the condition fails at the level of language—the truth may be inexpressible—and statistical dissonance is precisely the detector of that failure. Each successful refinement restores the grain on the path of play: post-restoration, some expressible theory matches the visited conditionals within tolerance. A SCUE is a configuration in which the grain of truth holds on-path for every agent and no one has a priced reason to look for the off-path remainder.*

Proposition 9.4 (Absorption; sketch). *Suppose tests are consistent (lemma 5.2), tolerances are generic, conjectures are empirically updated, and ties in equation (3) are broken by a fixed rule. Then almost surely there is a generation after which every agent's awareness is constant and the population is in one of two absorbing regimes:*

- (i) *a SCUE, in which every agent lives in a Bayesian world; or*
- (ii) *dissonant stasis for a nonempty set of agents: dissonance recurs but inquiry is never both elected and successful—because $\lambda_i B_i < \gamma_i$, or because elected production fails thereafter—and the affected agents carry forward committed policies with standing anomalies.*

If, whenever dissonance occurs with $B_i > 0$, the election threshold is met and ϕ_i succeeds with probability bounded away from zero, then regime (ii) with $B_i > 0$ is transient and absorption into a SCUE or into zero-leverage stasis is almost sure.

Proof sketch. Lemma 9.1 gives a terminal refinement almost surely. After it, awareness is fixed; lemma 5.2 implies each agent's active set is eventually constant almost surely. If it is eventually nonempty for all agents, posteriors converge on the finite theory space, empirical conjectures converge to induced play, and fixed tie-breaking makes policies eventually constant; the limit satisfies (i)–(iii), and (iv) holds because a met election threshold with success probability bounded below would produce a further refinement, contradicting terminality. If some agent's active set is eventually empty, dissonance recurs; terminality again forces the absence of successful elected production, which is regime (ii). The final claim iterates: each met-threshold dissonance episode produces a refinement with probability bounded below, and only finitely many refinements exist. A full treatment, including uniform integrability of the test thresholds across the finitely many awareness regimes, is deferred to the paper. $\square$

Remark 9.5 (Rational anomaly tolerance). *Zero-leverage dissonant stasis is not a pathology of the model but a recognizable condition: the agent knows its causal account is broken—every expressible theory is rejected—yet the break does not disturb any represented action comparison, so repairing it is worth nothing. Standing anomalies tolerated by working scientists and managers alike have exactly this structure. The model locates them: anomaly tolerance is rational precisely when pinned distinctions have no robust action leverage.*

Proposition 9.6 (On-path adequacy, and where error lives). *In a SCUE, every theory in the support of every μ_i is η -adequate at every visited pair. All surviving falsity is confined to what the configuration does not expose: mechanism assignments at states pooled with causally different states whose mixture imitates a single mechanism at the visited profiles, and counterfactual claims at unvisited profiles.*

Proof. Immediate from [definition 9.2\(iii\)](#) and [lemma 5.2](#); the residual clause is the complement of the visited pairs' constraints. $\square$

Proposition 9.7 (Persistent performance differences). *There exist primitives and absorbing configurations in which agents hold different terminal partitions and theories, expected per-generation payoffs differ strictly and permanently, and no disadvantaged agent has a priced repair: for each such agent either $B_i = 0$ or $\lambda_i B_i < \gamma_i$. The laboratory supplies instances: an agent with the correct factor refinement earns three-quarters against a coarse rival's one-quarter forever when the rival's inquiry cost exceeds its leverage; and under forgiving demand, a population sharing a false formulation splits the market at pooled actions indefinitely, each member half-warranted, none dissonant, none with positive leverage.*

Proof. The displayed instances are (the asymmetric-insight and trap configurations of [Ryall, 2026a](#)) embedded via [section 10](#); the cost-based instance is absorbing by [proposition 9.4\(ii\)](#) and the trap instance satisfies [definition 9.2](#) directly at the forgiving-demand parameter. $\square$

The persistence mechanism is the point: performance differences persist not because an equilibrium concept freezes them but because the process that would erase them has died economically—the losing configuration generates no dissonance, or generates no leverage, or prices the question below its cost. Heterogeneous terminal awareness is the general model's version of the laboratory's path dependence, and its comparative statics run through the two sources of diagnosticity: strategic variety inside one's cells and the environment's refusal to pay false theories quietly.

Definition 9.8 (Diagnostic environments). *The primitives (Θ, τ, ρ, u) are diagnostic if under every profile sustainable in a SCUE, every expressible theory that is payoff-relevantly false—false at some state where the truth would change the holder's optimal action—is η -inadequate.*

Proposition 9.9 (No payoff-relevant error in diagnostic environments). *In a diagnostic environment, every SCUE is free of payoff-relevant error: value-relevant falsity survives only in environments that admit concealing profiles. The laboratory's demand parameter is the sharpest instance: its $\beta < 1$ environments are diagnostic and its $\beta = 1$ environment is not.*

Proof. By [definition 9.2\(iii\)](#), SCUE theories are not η -inadequate; in a diagnostic environment payoff-relevantly false theories are, so none survives. The laboratory instances are verified in ([Ryall, 2026a](#)). $\square$

10 The four-state laboratory as an instance

General object	Laboratory instance
Agents N	Two firms.
States S	Customer states $\{x_{00}, x_{01}, x_{10}, x_{11}\}$ (written x there, s here).
Actions A_i	Architectures $\{0, 1\}$.
Consequences x_i	Own payoff together with the observed action profile.
Mechanisms Θ	Labels $\{0, 1, n\}$: $\theta^a(a_i, a_j)$ pays the winner-take-all/market-splitting payoffs under preferred architecture a ; θ^n pays one-half throughout. The demand parameter β rescales the pooled-misservice payoffs.
Objective assignment τ	The row law d : $\tau(s_{rc}) = \theta^r$.
Test	$\eta = 0, \bar{n} = 1, w \equiv 0$: exact consistency.
Dissonance	Empty active set after the diagnostic batch.
Coherence	Conflict-free cells; incoherence is pairwise, so coloring a graph.
Complexity κ	Coordinate complexity; the factor-seeking filter.
Pinned distinctions	Separation of the two conflicting labeled tokens.
Leverage B_i	$ L_i /2$, exact at $\beta = 1$.
Reflective assessment	$\mathcal{T} = \{d^r, d^c\}$, symmetric v ; warrant one-half, gap one-half.
Concealing profile	Shared column policy at $\beta = 1$ (the trap).
Diagnostic environment	$\beta < 1$.
SCUE instances	The trap ($\beta = 1$); the all-correct configurations; the asymmetric cost-based stasis.

The laboratory is thus a two-agent, four-state, deterministic instance of every object above, and its named results are instances of the general statements: revelation and demand-side revelation instantiate exposure; conflict-free restoration instantiates cellwise coherence; the row–column theorem instantiates [theorem 6.5](#); complete awareness instantiates the join of [proposition 6.6](#); and the trap, zero-leverage, and permanent-gap results instantiate concealment, [proposition 7.3](#), and [theorem 8.2](#).

11 Cognitive interpretation and checklist

The mapping to the account of inquiry in [Lonergan \(1992\)](#) and [Ryall and Wilkins \(2018\)](#) carries over from the laboratory with one addition made possible by the stochastic setting. Experience is the flow of episodes; the question for intelligence is dissonance-localized; heuristic anticipation is restoration, minimality, and the complexity ordering; insight’s product is the refinement; formulation is the mechanism assignment on the new cells; and decision is the committed policy. The addition concerns judgment. In the deterministic laboratory, judgment could be identified with fresh diagnostic labels; here, [remark 4.2](#) shows that within-language updating can never substitute for it—the posterior ranks expressible answers but cannot ask whether any expressible answer is adequate. Judgment is implemented by the consistency test and by warrant accounting ([theorem 8.2](#)), faculties formally distinct from the posterior, which is the model’s rendering of the distinction between

understanding and reflective judgment. The election ε_i remains the economic act that prosecutes an unrestricted desire to know under restricted resources.

Against the no-baked-in checklist: agents hold no prior over the outputs of ϕ_i (definition 6.3); questions are generated by defects, not drawn from a primitive list (definition 5.3); inquiry can fail or produce false formulations (assumption 6.4, theorem 6.5); leverage bounds are subjective and coarse (definition 7.2); identification respects observational equivalence (theorem 6.5); pre-insight objects are $\mathcal{P}_i$ -measurable (definition 3.4); no discovery supergame is optimized (proposition 7.3 prices only the event of success); the generating record is separated from warrant (theorem 8.2); rival behavior is endogenous (assumption 3.6); and the representational primitive is stated: Θ , the recoding capacity, and κ .

12 Open items

The note leaves for the paper, in rough order of importance: the full absorption theorem behind proposition 9.4; comparative statics of terminal heterogeneity in the two diagnosticity channels; equilibrium analysis where value destruction creates anti-coordination incentives; characterization of counterfactual transparency and its effect on the suppression of inquiry (remark 7.4); endogenous diagnosticity, where the demand side responds to misservice; test design as a choice variable; Markov state processes; and the deeper dimension of awareness growth in which the mechanism vocabulary Θ itself expands.

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